

Phantom (m:phant) Spec Showcase

Expression: a + [phantom(xyz)] + b

Case 1: No m:phantPr (bare m:phant)

Case 2: Empty m:phantPr

Case 3:

1. Definite integral (baseline — msubsup):

$$\int_{\square}^{\square}()$$

2. Word's indefinite integral (uses subHide/supHide m:val=1):

$$\int()$$

3. Summation default limLoc — Finding A trigger:

$$\sum_{\square}^{\square}\square$$

4. Summation explicit limLoc=subSup (control, different vars):

$$\sum_{\square}^{\square}\square$$

5. Product default limLoc — Finding A:

$$\prod_{\square}^{\square}\square$$

6. Union default limLoc — Finding A (also supHide from Word):

$$\bigcup_{\square}^{\square}\square_{\square}$$

2b. Crafted indefinite integral — NO m:sub/m:sup elements (Finding C real trigger):

$$\int_{\square}^{\square}\square$$

7. subHide m:val="true" (Finding B — anchors commit 2bd58d3):

$\int \square$

8. Bare **subHide** (Finding B — spec bare-flag=ON):

$\int \square$

9. **limLoc** present-no-val (Finding A — spec default=undOvr):

$\sum \square$

10. **chr** present-no-val (Finding D — empty character):

$\int \square$

~~12. **subHide** m:showval="0" (ST_OnOff OFF value — §22.9.2.7): sub should stay visible → msubsup~~

Case 4: m:show val=1 (visible)

Case 5: m:show val=0 (hidden)

Case 6: m:show val=true

Case 7: m:show val=false (hidden)

Case 8: zeroWid alone

Case 9: zeroAsc alone

Case 10: zeroDesc alone (classic smash)

Case 11: show=0 + zeroDesc=1 (hidden, smashed)

Case 12: all three zeroed, show absent

Case 13: all three zeroed + show=0

Case 14: m:transp=1 (unsupported)



13. m:grow m:val="0" (Finding E — §22.1.2.72): operator should NOT grow (expect largeop/stretchy=false on mo)

